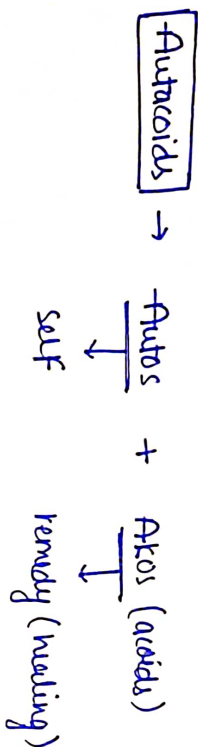


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UNIT - IIIrd

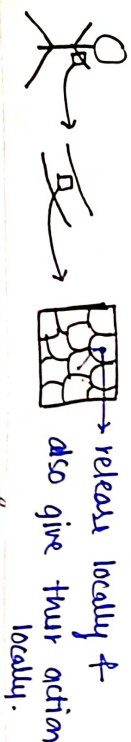
AUTACOIDS AND RELATED DRUGS



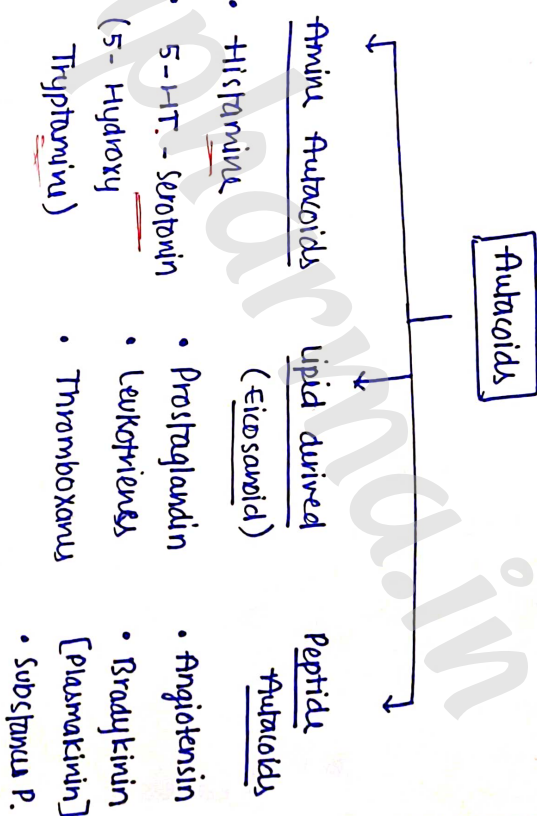
- The chemical substances produced within the body to help in the remedy of local injuries.

- also known as **Autacoids**.

- These are those chemical substance which act as the local hormones, means release from the tissue and act at the site of synthesis and release.



- Classification of Autacoids



Others → cytokines, gasch, Somatostatin, endothelins etc.

③ Point to be cover :-

- **General Introduction**
- Synthesis, Storage, Release & Metabolism
- Receptor
- **Pharmacology role (action)** → CNS, CVS, GIT, Blood vessels etc.
- **Antagonist (if asked)** → their drugs.

- Synthesis →

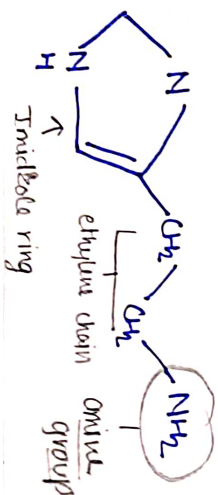
- Histamine $\rightarrow$ Hist + amine
 $\downarrow$
 Tissue NH₂ group

- If release from an local tissue (cells) and contain an amine group.

- If play an major role in many allergic reactions such as sneezing etc.

— Histamine is discovered by Sir Henry Dale in 1910.

- structure →
of histamine



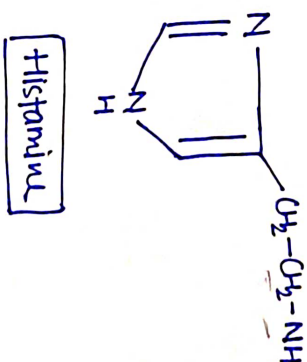
NC(=O)C1=CN=CN=C1

Histidine

Histidine Decarboxylase

$(-CO_2)$

- Histamine is synthesized by decarboxylation of histidine, which is found in all organs and tissue of the human body.



- Storage →

- Histamine mostly stored (present) in Mast cells and basophils.

- Also stored in enterochromaffin-like cells (ECL) in GIT and variety of neurons. (histaminergic)

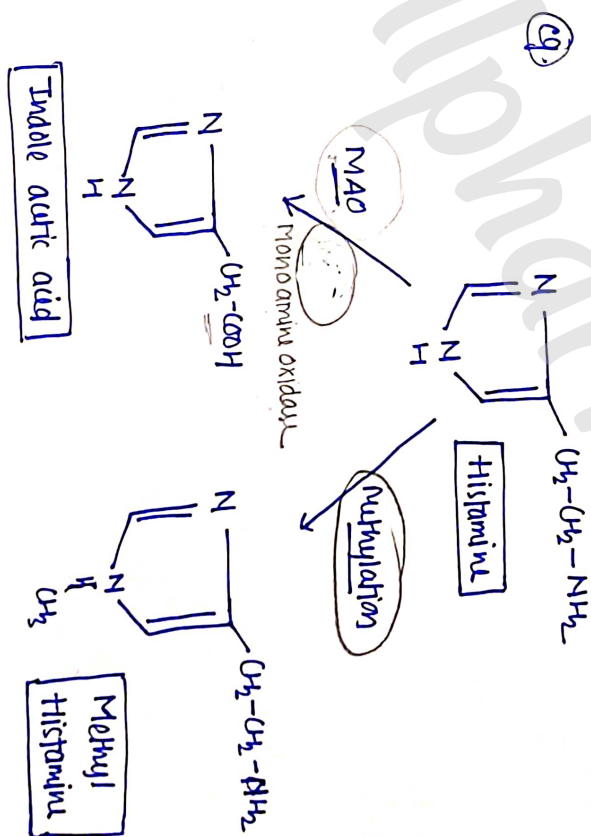
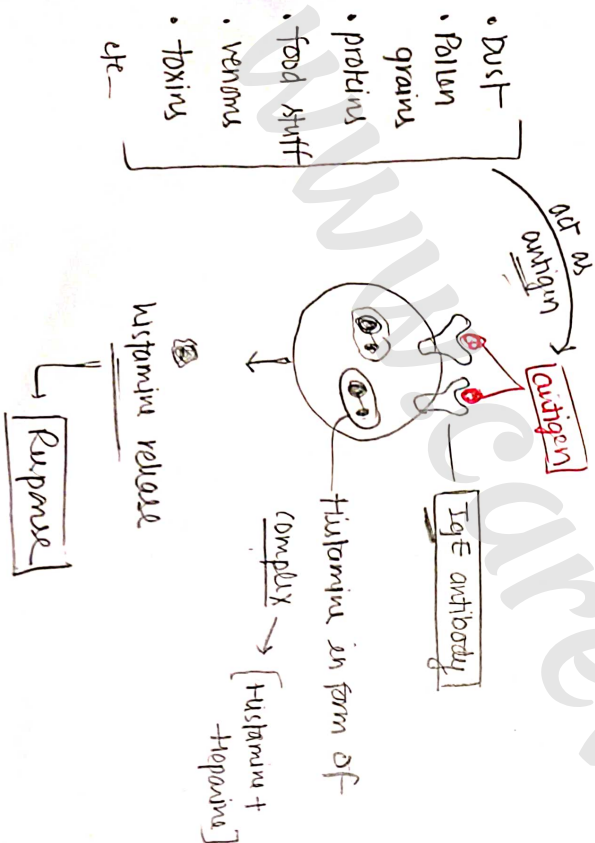
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- Usually histamine (mast cells) found in all over the body, but in high concentration it presents in skin, mucosal cells of lungs, intestine, urinary tract etc.
- Catabolism → Process of inactivation histamine after use which change the structure of histamine and inactivate it.

• Release →

It released by the interaction of antigen with IgE antibodies on the surface of mast cells. (all over the body).



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• Histamine Receptor →

There are mainly four receptor, in which histamine bind and give their pharmacological effects :-

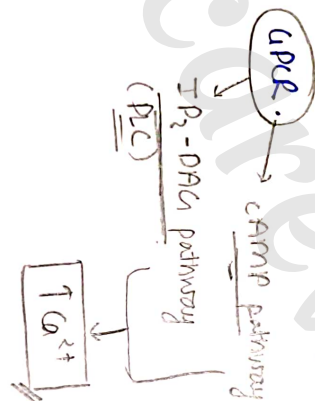
H_1

H_2

H_3

H_4

All these receptors are G-protein couple receptor (GPCR) and give their action by the mechanism of GPCR.



Physiological role

RECEPTOR	LOCATION	PATH	EFFECTS (ACTION)	EXAMPLES
H_1	CNS, GI Tract, Bronchi, Heart	$\uparrow IP_3$ -DAG Pathway	- Bronchoconstriction, - Pain, Pruritus (itching) - Vasodilation, Inflammation etc..	Histamine, 2-Methyl histamine
H_2	Gastric parietal cells, CNS, Heart and Vascular smooth muscle.	$\uparrow cAMP$	$\uparrow$ Gastric acid secretion ($\uparrow HCl$) & vasodilation	Histamine, 4-methyl histamine
H_3	CNS (mostly in presynaptic neurons)	$\downarrow cAMP$	Inhibition of histamine release	Histamine, Methyl histamine
H_4	Hematopoietic cells (stem cells that give rise to other blood cells).	$\uparrow cAMP$	- Modulate immune function - Inflammation & allergic reaction	

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• Pharmacological role (action) of Histamine

④ Inflammation & some allergic reactions

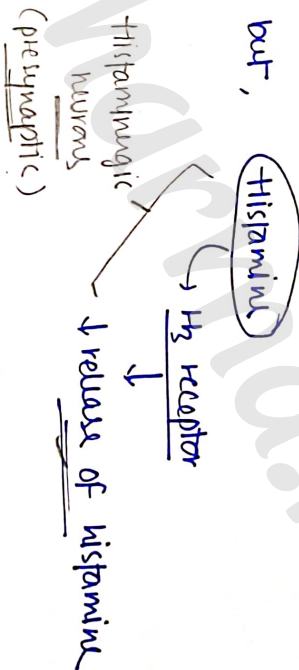
① Smooth muscle of

respiratory tract → It cause bronchoconstriction

cause
Asthma

⑤ CNS → Not well understood

but,



② CVS (cardiovascular system) →

Vasodilation
in blood vessels

Heart → stimulation

↑ force of contraction

by H₂ receptor

③ GI tract →

Histamine

H₂ receptor

↑ gastric secretion

from parietal cells

↑ HCl

• Antagonist of histamine (Antihistamines)

• These are those agents or drugs which inhibit the action of histamine

• Basically there are of two types:—

H₁-Antagonist and H₂-Antagonist

• They give their effect by blocking the H₁ & H₂ receptor so histamine does not bind & not give any action. eg.

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① H₁ antagonists →

- Diphenhydramine
 - Triphenylamine
 - Triprolidine
 - Promethazine
- } first generation
- Cetirizine
 - Levocetirizine
- } second generation
- etc...

② H₂ antagonists →

- cimetidine
- famotidine
- Ranitidine

* for further more details

refer Antihistaminic agents, Medicinal chemistry 2

5th Sem

- effects (pharmacological rxn) →
oppose the effects of histamines.

5-Hydroxytryptamine [5-HT]

It is also known as [serotonin].

- It is also belongs to an amino acid which release in our body.
 - It is derived from an amino acid and mainly found in AIT, platelets and CNS.
- * Action →

- It regulate our mood, behaviour, sleep, depression, pain, sexual activity, muscle contraction and intestinal movement etc..

• Biosynthesis →

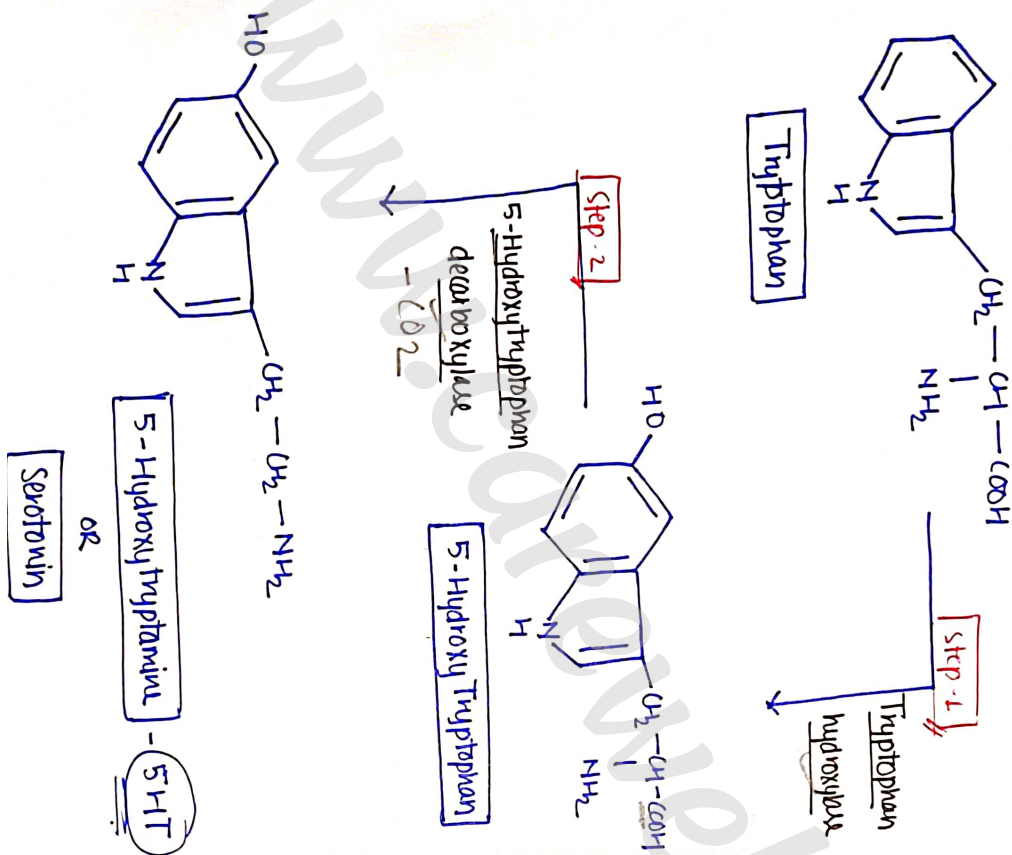
It involves two steps:—

- ① Tryptophan (amino acid) is converted into 5-hydroxytryptophan (5-HTP) with the help of enzyme Tryptophan hydroxylase.
- ② Now this 5-hydroxytryptophan (5-HTP) is

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Converted into serotonin (5-Hydroxytryptamine) with the help of enzyme decarboxylase



Storage →

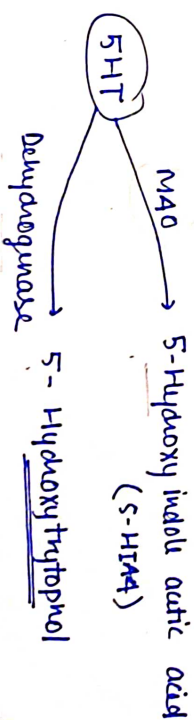
They stored inside granules (vesicles) in

- enterochromaffin cells of GIT (mostly about 90-95%).
 - Serotonergic neurons in CNS
 - Platelets
- Now, serotonin is synthesized in both enterochromaffin cells + serotonergic neurons but not in platelets (decarboxylase enzyme absent), they only store it.

Also they release from these storage sites.

Degradation →

The effect of serotonin is terminated by the enzyme MAO and Dehydrogenase which change their structure. so they do not bind with receptor and not give any action.



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• Serotonin (5-HT) Receptor →

There are basically seven types of receptor, which also have their subtypes.

- ① 5-HT₁ → 5-HT_{1A}, 1B, 1D, 1E, 1F, 1G
- ② 5-HT₂ → 5-HT_{2A}, 2B, 2C
- ③ 5-HT₃
- ④ 5-HT₄
- ⑤ 5-HT₅ → 5-HT_{5A}, 5-HT_{5B}
- ⑥ 5-HT₆
- ⑦ 5-HT₇

- All receptor except 5-HT₃ are G-PCR and follow the mechanism by two pathways cAMP or IP₃-DAG.

- 5-HT₂ are ligand gated ion channel.

There are four major receptor on which serotonic give their pharmacological role.
5-HT₁, 5-HT₂, 5-HT₃, 5-HT₄

Receptor	Location	Path	Pharmacological Action (Role) of Serotonin	Agonist	Antagonist
5-HT ₁	CNS, PNS	↓ cAMP (inhibitory effects)	• ↓ release of neurotransmitter (5-HT) • Hyperpolarisation (activate K⁺ channel) • facilitate Adrenaline + NA release etc..	• Buspirona	Ergotamine Methylsergid
5-HT ₂	Smooth muscles (GIT), Platelets, CNS - cerebral cortex	↑ IP ₃ -DAG (PLC activation)	• Smooth muscle contraction (intestine) • Bronchoconstriction, Platelet aggregation • 5-HT_{2A} → vasoconstriction • 5-HT_{2C} → vasodilation ([EDRF] release).	• α-methyl 5-HT	• Cypro-heptadine • Ketaserin
5-HT ₃	CNS, PNS	Activated ion channel	• stimulated - pain, itching, Nausea, Vomiting • Bradycardia (↓ BP) • Emetic reflex	• 2-methyl 5-HT	• Ondansetron • Dolasetron
5-HT ₄	CNS - Hippocampus GIT	↑ cAMP	• ↑ intestinal secretion and ↑ peristalsis movement • Neuronal excitation.	• Cisapride • Renzapride	

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• Pharmacological Role (Action)

① CVS (CardioVascular system) →

- $5HT_2A \rightarrow$ Vasoconstriction
- $5HT_{2E} \rightarrow$ Vasodilation
- Platelet aggregation
- Bradycardia ($\downarrow$ B.P.)
- Stimulated heart $\rightarrow$ (+ve inotropic effects by release of $(NH_4)^+$ ~~ATP~~)

② CNS →

- $\downarrow$ release of neurotransmitter (5-HT)
- Sleepiness
- Pain perception

③ Smooth muscles →

- Respiratory system $\rightarrow$ bronchoconstriction
- $\uparrow$ peristalsis movement

④ Glands →

- $\downarrow$ gastric secretion

⑤ Platelets →

$\uparrow$ aggregation of platelets

5-HT Antagonist

These are those drugs which inhibit the action of 5-HT (serotonin).

Drugs →

- ~~Effet~~ • Cyproheptadine
- Methysergide
- Ketanserin
- Clozapine
- Risperidone
- Ondansetron etc

① Cyproheptadine →

- Blocks $5-HT_{2A}$ receptor.
- Also has H_1 -Antagonist, anticholinergic, and sedative properties.

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• Pharmacological Role (action)

① CVS (CardioVascular System) →

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- ~~type~~ • Cyproheptadine
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- Ketanserin
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Uses →

• ↑ appetite

• also used to cardioid syndrome

Adr →

• Orowsinus

• Dry mouth

• Confusion

• weight gain

②

Mefenorexide →

• Blocks $5-HT_{2A/2C}$ selectively

Uses :-

• migraine

• carcinoid

Adr →

• Nausea, Diarrhea, Abdominal pain

③

Oronasection →

• Blocks $5-HT_3$

Uses →

to control nausea and vomiting

Adr → Headache, Mild constipation, Diarrhea

④ Ketan Serin →

• selective $5-HT_2$ receptor blockers.

Uses →

• effective antihypertensive

• employed in Raynaud's syndrome

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CHAPTER-3

PROSTAGLANDINS, THROMBOXANES AND

LEUKOTRIENES

② Ficoids → Fica - 20 carbon compounds

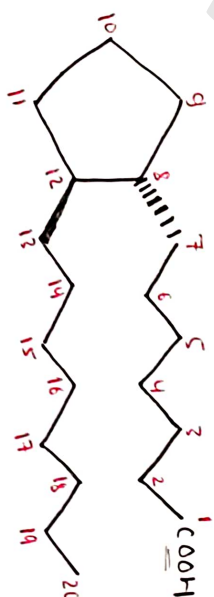
- All three compounds are produced from 0 thiocardonic acid (TIA) which is a 20 carbon compounds (unsaturated).

- Synthesis →

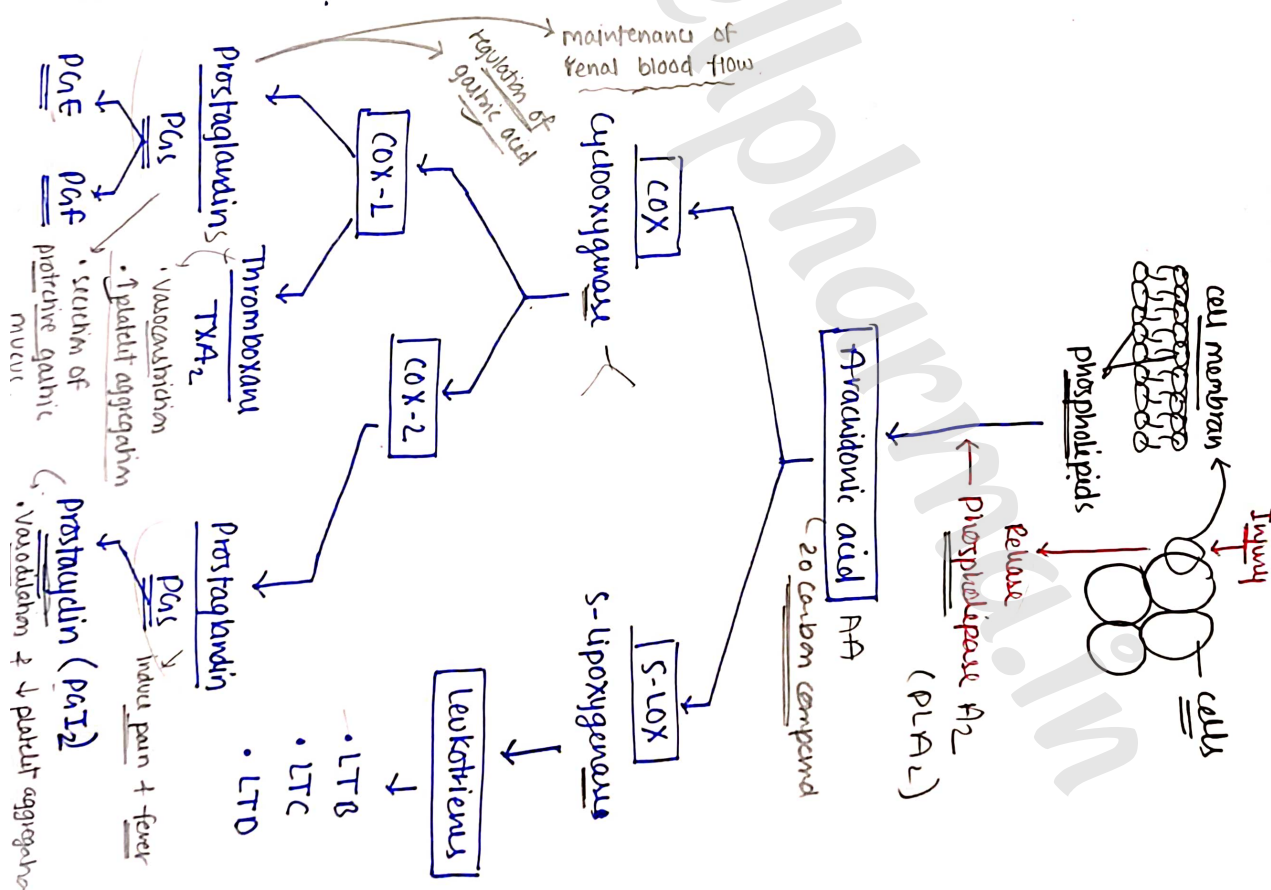
Chemically prostaglandins (PGs) are

Protanedic acid derivatives : It has

five membered ring and two side chain.



Prostanoic acid



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- During any type of injury, phospholipase A_2 (PLA₂) enzyme is released which further convert or hydrolyzed phospholipid into Arachidonic acid (AA).
- Now, this Arachidonic acid produce (release) two enzyme COX (cyclooxygenase) and 5-LOX (5-lipoxygenase) or we can also say two pathways.
- Now, COX enzyme produce Prostaglandins and Thromboxane and 5-LOX enzyme produce leukotrienes.
- Prostaglandins $\rightarrow$ P_Gs, P_GE, P_GI, P_GF etc.
- Thromboxane $\rightarrow$ TXA₂
- Leukotrienes $\rightarrow$ LTB₂, LTC₄, LTD₄ etc.
- Prostaglandins (P_Gs) $\rightarrow$ It is a group of compounds derived from unsaturated 20-carbon fatty acids, primarily Arachidonic acid via COX pathway.
- The name prostaglandins (P_Gs) was derived from the gland from these compound were first isolated.
- P_Gs act on its receptor and show their effect.
- Thromboxane (TXA₂) $\rightarrow$
 - It is a substance which is made by platelets that cause blood clotting (platelet aggregation) and vasoconstriction.
 - It is also derived from arachidonic acid and are related to prostaglandins.
- Prostaglandin (PGI₂) $\rightarrow$
 - Derived as same as other but it cause vasodilation and also it inhibit the platelet activation oppose thromboxane.

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• Pharmacological Action (Role) of

Prostaglandins, Thromboxans & Prostacyclin
 PGI_1 TXA_2 PGI_2

① CVS →

- PGI_2 → cause vasodilation + fall in BP ($\downarrow BP$)
- PGI_2 → constricts many veins including pulmonary vein + artery
- PGI_2 → uniformly vasodilatory effect + more hypotensive than PGI_1 .
- TXA_2 → vasoconstriction + $\uparrow$ platelet aggregation.

② GIT →

- PGI_2 → reduce acid secretion in stomach also $\downarrow$ pepsin.
- protective effect on gastric mucosa → $\uparrow$ secretion of mucosa, water + mucus secretion.
- PGI_2 → same but less potent.

• PGI_1 → $\uparrow$ peristalsis movement

③ Platelets →

- TXA_2 → potent vasoconstrictor + $\uparrow$ platelet aggregation
 - PGI_2 → potent vasodilator + $\downarrow$ platelet activation.
- (guaranteed by vascular endothelium)*

④ Bronchial muscles →

- PGI_2
 - PGD_2
 - TXA_2
- bronchoconstrictor (potent) (Asthma)

- PGI_2 → bronchodilator (potent)
- PGI_2 → bronchodilation (mild)

not used, because they produce irritation

$\downarrow$ IOp - intraocular pressure in eye



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⑤ Uterus →

- PGI_2 } contraction during or non-during pregnancy
- $PGF_{2\alpha}$

⑥ Kidney →

- PGI_2 } renal vasodilation & inhibit tubular reabsorption.
- PGI_2
- TXA_2 → causes vasoconstriction.
- PGI_2 , PGI_2 + PGD_2 } evoke release of renin. (continues)
- ⑦ CNS →
- PGI_2 → sedation, rigidity, behavioural changes
- PGI_2 → induce fever
- They also produce pain.

Adrs →

Pain, Nausea, Vomiting, diarrhoea, vaginal bleeding, flushing, shivering, fever, fall in BP etc—

⑧ Leukotrienes → (LT)

It is also an eicosanoid and it produced from arachidonic acid.

Types → LTB_4 , LT_4 + LTD_4

— Pharmacological role (action) →

① CVS →

- LT_4 } exudation of plasma.
- LTD_4 } secretion → ↑ capillary permeability
- They may ~~facilitate~~ facilitate local release of histamine from mast cells.
- ↓
- chronic inflammatory states.

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② Smooth muscle →

• LT_{C4}
• LT_{D4} } Bronchoconstriction, ↑ mucus
secretion in
airway

③ Afferent nerve →

• ↑ pain impulse
• inflammation

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Table 13.1 : A summary of the actions of major prostaglandins, prostacyclin and thromboxane

Organ	Prostaglandin E_2 (PGE_2)	Prostaglandin $F_{2\alpha}$ ($PGF_{2\alpha}$)	Prostacyclin (PGI_2)	Thromboxane A_2 (TXA_2)
1. Blood vessels	Vasodilatation, ↓ BP	Constricts larger veins and some arteries, little effect on BP	Vasodilatation (marked and widespread), ↓ ↓ BP	Vasoconstriction
2. Heart	Weak inotropic, reflex cardiac stimulation	Weak inotropic	—	—
3. Platelets	Variable effect	—	Antiaggregatory	Aggregation and release reaction
4. Uterus	Contraction (<i>in vivo</i>), softening of cervix	Contraction (<i>in vivo</i> and <i>in vitro</i>), softening of cervix	—	Contraction (<i>in vitro</i>)
5. Bronchi	Dilatation, Inhibit histamine release	Constriction	Dilatation (mild), inhibit histamine release	Constriction
6. Stomach	↓ acid secretion, ↑ mucus production	—	↓ acid secretion (weak), mucosal vasodilatation	—
7. Intestine	Contracts longitudinal & relaxes circular muscles, ↑ peristalsis, ↑ Cl^- & water secretion	Spasmogenic, ↑ fluid & electrolyte secretion (weak)	Weak spasmogenic, inhibits toxin-induced fluid secretion	Weak spasmogenic
8. Kidney	Natriuresis, ↓ Cl^- reabsorption, inhibit ADH action, vasodilatation, renin release	—	Natriuresis, vasodilatation, renin release	Vasoconstriction
9. CNS	Pyrogenic, variety of effects on i.c.v. inj.	—	Same as PGF_2	—
10. Afferent nerves	Sensitize to noxious stimuli → tenderness	—	—	—
11. Endocrine system	Release of ant. pituitary hormones, steroids, insulin; TSH-like action	—	—	—
12. Metabolism	Antihypolytic, insulin like action, mobilization of bone Ca^{2+}	—	—	—

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NSAIDS

Non-Steroidal Anti Inflammatory Drugs

Anti-Inflammatory

oppose

inflammation

such as
pain, fever, allergic response,
Redness, swelling etc.

- These are those drugs which have

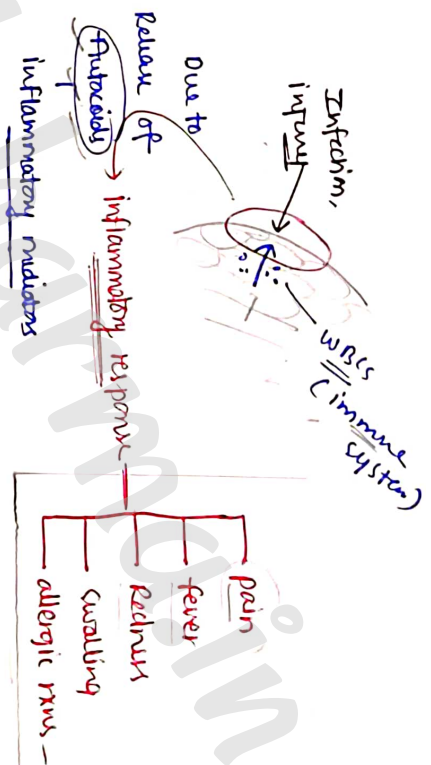
anti-inflammatory effects means they reduce inflammatory responses such as pain, fever etc.

OR

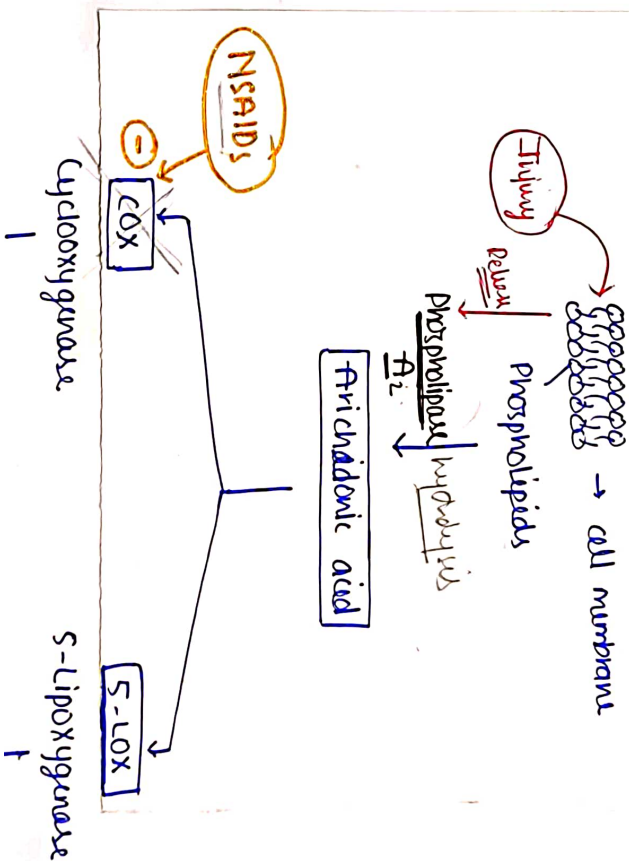
- These are those drugs which reduce pain or fever.

- Inflammation →

It is a process by which our body's white blood cells (immune system) protect our body from infection (foreign materials).

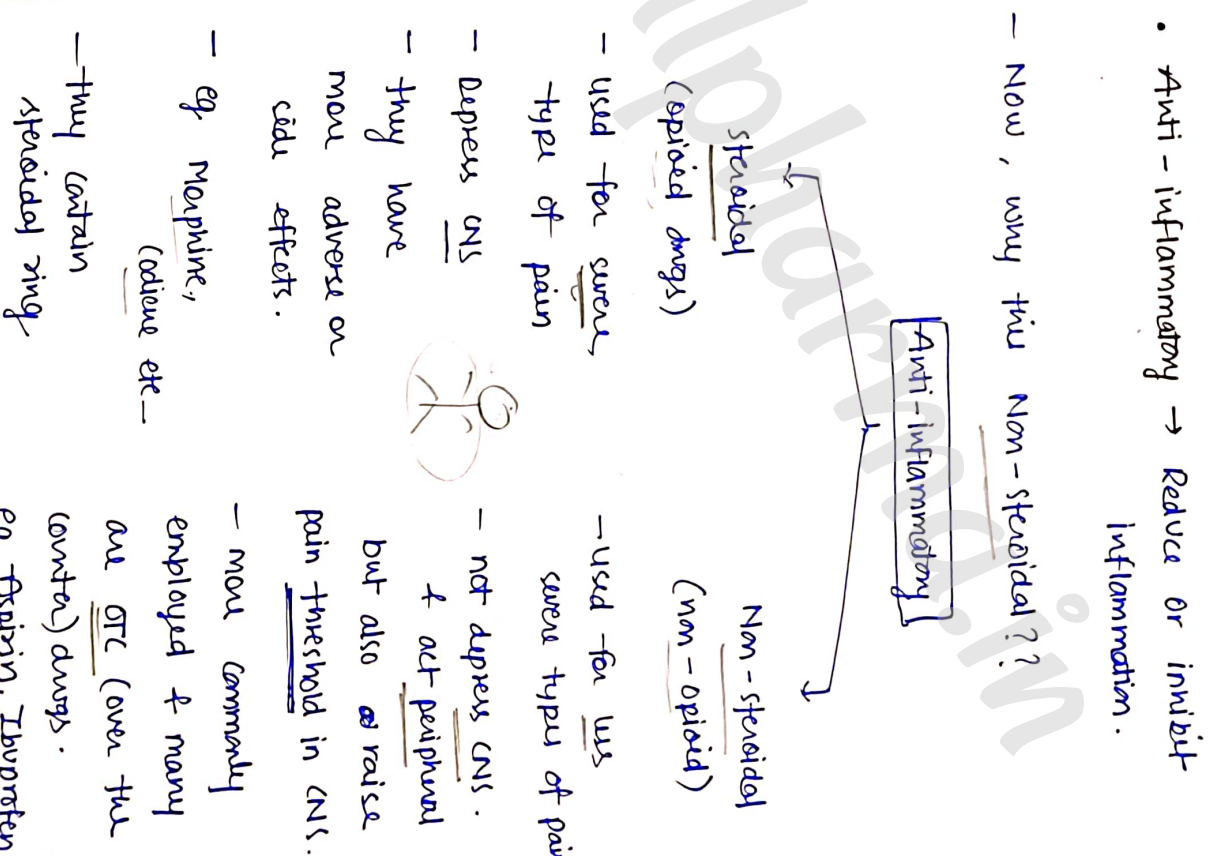
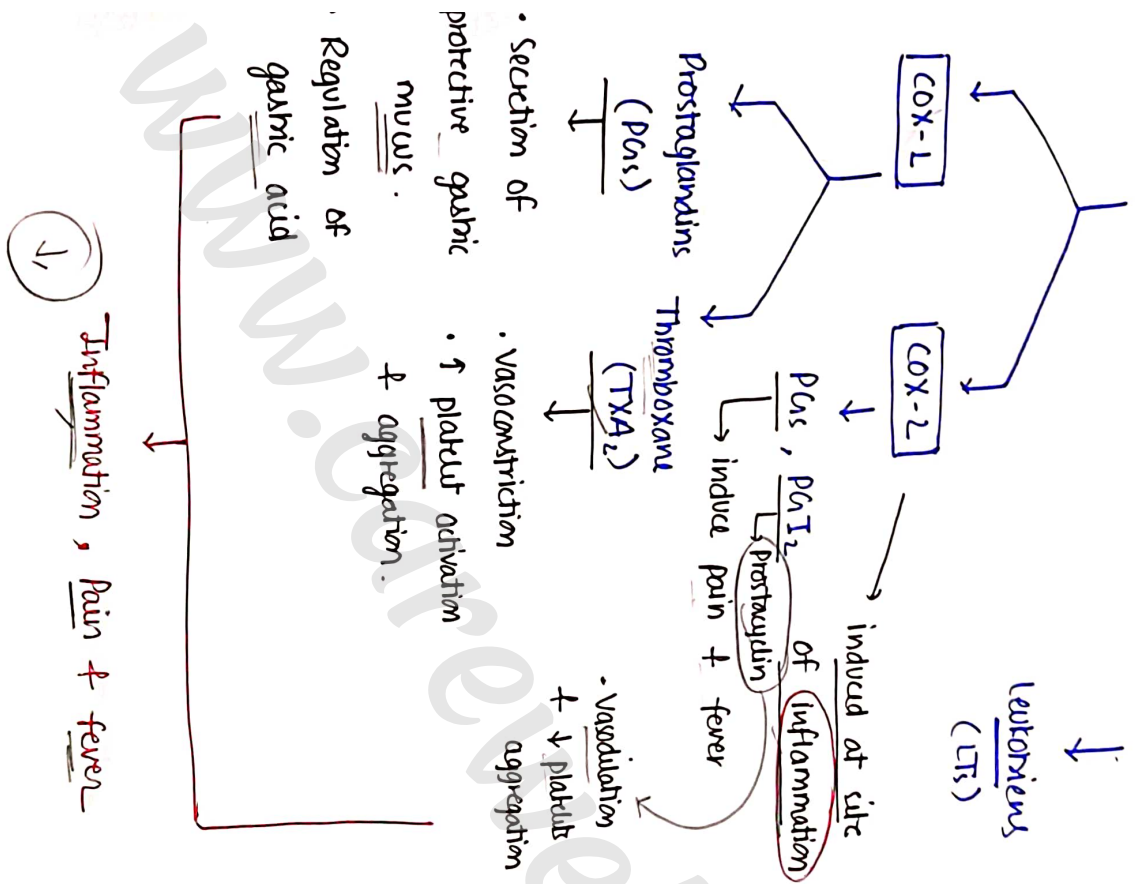


Mechanism of Inflammation & NSAIDS



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• NSAIDs →

Non-steroidal anti-inflammatory drugs.

Classification

- KD Tripathy

① Non-selective COX inhibitors

- ① Salicylates → Aspirin
- ② Propionic acid derivatives → Ibuprofen, Naproxen, ketoprofen, flurbiprofen
- ③ Fenamate → Mefenamic acid
- ④ Enolic acid derivatives → Piroxicam, Tenoxicam
- ⑤ Acetic acid derivatives → Ketorolac, Indomethacin, Nabumetone
- ⑥ Pyrazolone derivatives → Phenylbutazone, Oxyphenbutazone, Propyphenazone

② Preferential COX-2 inhibitors

Nimesulide, Diclofenac, Acetofenac,
Meloxicam, Etoxicolac

③ Selective COX-2 inhibitors

Celecoxib, Etoricoxib, Parcoxib

④

Analgesic - Antipyretics with poor
anti-inflammatory action

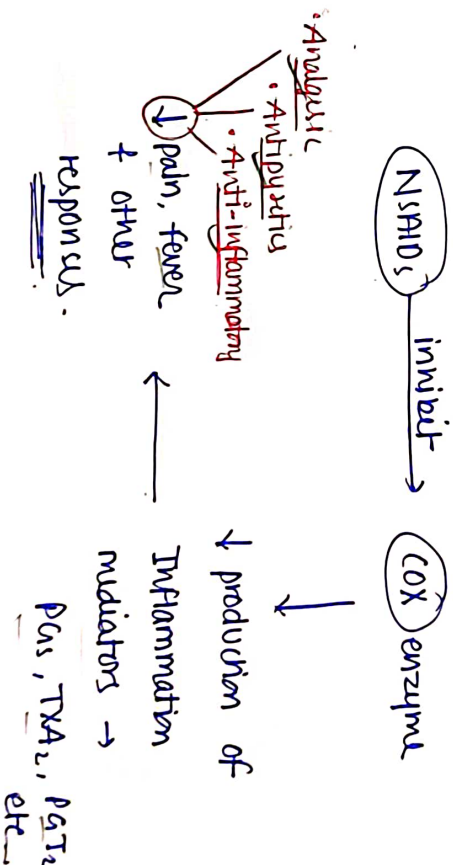
⑤ Para-aminophenol derivatives →

Paracetamol (Acetaminophen)

⑥ Benzoxazone derivatives →

Nefopam

⑦ General mechanism →



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① Non-selective COX inhibitors →

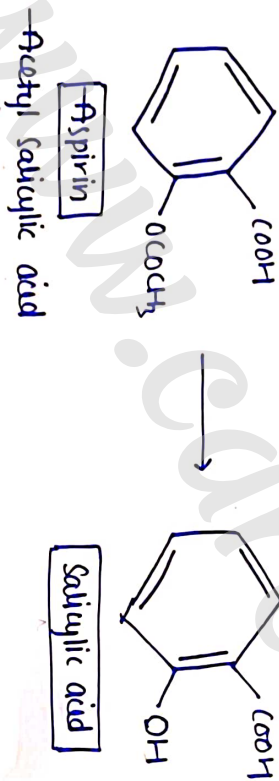
These are those drugs which inhibit the COX enzymes (both 1 & 2, but mostly 1st).

② ASPIRIN →

Also known as acetyl salicylic acid.

- It is a protype drugs.

• It firstly converted into salicylic acid in the body then it gives their action.



- MOA →

- It inhibits the enzyme COX
- which further decreases the production of

inflammatory mediators such as pro

- Due to the decrease inflammatory responses such as pain, fever etc --

- Pharmacological action →

- ① It gives analgesic, antipyretic & anti-inflammatory actions.

- ② GI → • irritate gastric mucosa and cause peptic ulcers.

- ↑ risk of GI bleeding
- cause Nausea, vomiting

- ③ Blood → Anti-platelets effect by inhibiting TXA₂.

- Uses →

- As analgesic, antipyretic, anti-inflammatory
- acute rheumatic fever & also antiplatelets.
- Rheumatoid arthritis & other inflammation.

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— pharmacokinetics →

⇒ Absorption → Through stomach & small intestine.

• Aspirin deacetylated → salicylic acid

⇒ Metabolism → Liver

• Slow entry in Brain (BBB)

• easily pass — placenta

⇒ excretion → urine

• plasma $t_{1/2}$ → 15-20 min

and it is dose dependent.

— Adverse effects →

• Nausea, Vomiting

• Hypersensitivity & idiosyncrasy

• ↑ risk of GI bleeding & peptic ulcer

• ↑ risk of bleeding

• ↑ risk of renal injury

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